Iterated compression

Question: Should we rule out compressors that would expand some texts instead of reducing them?

Answer: No. Any given (non trivial) compressor will expand some the texts given to it!

This paradoxical observation becomes obvious once one realizes that the compressor is an injective function within the set of all possible texts of size N or less: if $Z(T_1) = Z(T_2)$, then $T_1 = T_2$. So if some texts are compressed, some others must be expanded.

It is easy to observe that "zip" does not re-zip its own output to a shorter string. One can even verify that applying "zip" repeatedly even leads to an ever growing string. To do so, execute the program rzip.py on this file: Kolmogorov_complexity--Wikipedia.txt. The program keeps applying "zip" `n` times. For instance, for two iterations:

```
$ rzip.py 2 Kolmogorov_complexity--Wikipedia.txt
    Test for repeated zip

Iteration 0: length = 24552
Iteration 1: length = 8660
```

Iterated compression

1/1 point (ungraded)

Let Z represent "zip", and T our initial text. For which value of n do we get $Z^n\,(T)) > \text{length}\,(T)$?

- 172
- 692
- 1236 (selected)

Answer
Correct:
"Zip" is unable to compress its output any further. Moreover, it must add a piece of information to keep track of the fact that it has been applied. The size of the file resulting from n applications of Zip grows like n^2. With $n = 1236$, it is larger than the initial size.

Submit You have used 1 of 1 attempt

Answers are displayed within the problem